CIVE 207 — Crib Sheet

Final

Hooke's Law

$\sigma = E\varepsilon \qquad \tau = G\gamma$

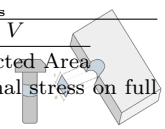
$\sigma = \dfrac{F}{A} \qquad \delta = \dfrac{PL}{AE}$

$\tau = \dfrac{V}{A}$ (single shear)

$\tau = \dfrac{V}{2A}$ (double shear)

Bearing Stress

$$\sigma_B = \frac{V}{\text{Projected Area}}$$



σ_{gross}: normal stress on full area

σ_{net}: stress at smallest area

Stress on Oblique Plane

$\sigma = \dfrac{P}{A}\cos^2\theta$

$\tau = \dfrac{P}{A}\sin\theta\cos\theta$

$\sigma_{\max}$: $\theta = 0°$

$\tau_{\max}$: $\theta = \pm 45°$

Factor of Safety

$$FS = \frac{\sigma_{\text{ultimate}}}{\sigma_{\text{allowable}}}$$

Shear Strain

$\gamma > 0$: inside angle decreases

$$\gamma = \theta_{\text{initial}} - \theta_{\text{final}}$$

Thermal Deformation

$$\delta = \alpha \cdot \Delta T \cdot L$$

3D Stress–Strain

$E\varepsilon_x = \sigma_x - \nu(\sigma_y + \sigma_z)$

$E\varepsilon_y = \sigma_y - \nu(\sigma_x + \sigma_z)$

$E\varepsilon_z = \sigma_z - \nu(\sigma_x + \sigma_y)$

Poisson Ratio

$$\nu = -\frac{\varepsilon_{\text{lateral}}}{\varepsilon_{\text{axial}}}$$

Dilatation (Unit Volume Change)

$e = \varepsilon_x + \varepsilon_y + \varepsilon_z$

$v = (1+\varepsilon_x)(1+\varepsilon_y)(1+\varepsilon_z) \approx 1 + e$

$e = v - 1$

e: unit change in volume

$$e = \frac{1-2\nu}{E}(\sigma_x + \sigma_y + \sigma_z)$$

Hydrostatic Pressure p

$$e = -\frac{3(1-2\nu)}{E}p$$

Bulk Modulus / Modulus of Compression

$$k = \frac{E}{3(1-2\nu)}, \qquad e = -\frac{p}{k}$$

$$G = \frac{E}{2(1+\nu)}$$

Saint-Venant Principle

At distance $\geq$ width of member, stress distribution is uniform whether plate-loaded or point-loaded.

Stress Concentration

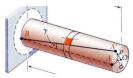
$$K = \frac{\sigma_{\max}}{\sigma_{\text{avg}}}$$

Torsion:

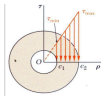
$$\tau_{\max} = K\left(\frac{Tc}{J}\right)_{\text{smaller shaft}}$$

Torsion & Shear Strain

$\gamma = \dfrac{\rho\phi}{L}$



$\tau = \dfrac{T\rho}{J}$



$\phi = \dfrac{TL}{GJ}$

Polar Moment of Inertia

$$J_{\text{circle}} = \frac{\pi}{2}r^4 = \frac{\pi}{32}d^4$$

Gear Ratios

$$\frac{T_1}{R_1} = \frac{T_2}{R_2}, \qquad \phi_1 R_1 = \phi_2 R_2$$

Power

$$P = T\omega = 2\pi f T$$

$$1\ \text{W} = 1\ \frac{\text{N}\cdot\text{m}}{\text{s}}$$

$$1\ \text{hp} = 550\ \frac{\text{ft}\cdot\text{lb}}{\text{s}}$$

Variable Cross-Section

$$\phi = \int_0^L \frac{T}{GJ}\,dx$$

$$\phi_{E/B} = \phi_E - \phi_B$$

(E moves w.r.t. B)

Modulus of Rupture in Torsion

$$R_T = \frac{T_U c}{J}$$

Plastic Torsion

Not Yield

$$\tau = \frac{T\rho}{J}$$

Yield

$$T_Y = \frac{J}{c}\tau_y$$

$0 \leq \rho \leq \rho_Y$: $\quad \tau = \dfrac{\tau_Y}{\rho_Y}\rho$

$\rho_Y \leq \rho \leq c$: $\quad \tau = \tau_Y$

$$T = \frac{4}{3}T_Y\left(1 - \frac{1}{4}\frac{\rho_Y^3}{c^3}\right)$$

$$T = \frac{4}{3}T_Y\left(1 - \frac{1}{4}\frac{\rho_Y^3}{c^3}\right)$$

$$T_P = \frac{4}{3}T_Y$$

$\dfrac{\rho_Y}{c} = \dfrac{\phi_Y}{\phi}$ (ϕ_Y = angle when outer starts yielding)

Torsion — Rectangular Cross-Section

$$\tau_{\max} = \frac{T}{c_1 ab^2} \qquad \phi = \frac{TL}{c_2 ab^3 G}$$

Hollow Cylinder (Elasto-Plastic)

$$T = \iint \rho\,\tau\,dA = 2\pi\int_a^b \tau\,\rho^2\,d\rho$$

$$T_e = \frac{\tau_{\max} J}{c}$$

$$T_{e\text{-}p} = 2\pi\tau_Y\left(\frac{b^3}{3} - \frac{a^4}{4\rho_Y} - \frac{\rho_Y^3}{12}\right)$$

$$T_p = \frac{2}{3}\pi\tau_Y(b^3 - a^3)$$

Residual Shear Strain

$$\phi_r = \phi - \phi' = \frac{\gamma_Y L}{\rho_Y} - \frac{TL}{JG}, G = \frac{\tau_Y}{\gamma_Y}$$

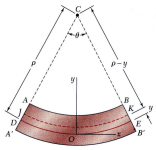
Thin-Walled Hollow Shafts

$$\tau = \frac{T}{2t\mathcal{A}}, \qquad q = \tau t$$

$$\phi = \frac{TL}{4\mathcal{A}^2 G}\oint\frac{ds}{t} \qquad \left(\sum_{i=1}^{n}\frac{s_i}{t_i}\right)$$

$\mathcal{A}$: area enclosed by centerline

Pure Bending



$$\varepsilon_x = -\frac{y}{\rho}, \varepsilon_{\max} = \frac{c}{\rho}$$

$$\sigma_x = -\frac{y}{c}E\varepsilon_{\max} = -\frac{y}{c}\sigma_{\max}$$

Neutral axis must pass through the centroid.

$$\sigma_x = -\frac{My}{I}, \sigma_{\max} = \frac{Mc}{I} = \frac{M}{S}$$

Moment of Inertia

Rectangle: $I = \dfrac{1}{12}bh^3$

Circle: $I = \dfrac{\pi}{4}r^4$

Parallel axis: $I = I_{CM} + Ad^2$

$S = \dfrac{I}{c}$ — section modulus

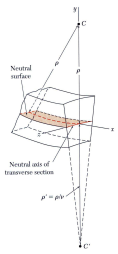
Rectangle: $S = \dfrac{1}{6}bh^2$

Curvature

$$\frac{1}{\rho} = \frac{\varepsilon_{\max}}{c} = \frac{\sigma_{\max}}{Ec} = \frac{M}{EI}$$

Anticlastic Curvature ρ'

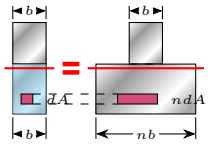
$$\varepsilon_y = \varepsilon_z = -\nu\varepsilon_x = \frac{\nu y}{\rho}, \frac{1}{\rho'} = \frac{\nu}{\rho}$$



Composite Materials

Replace higher-E material with low-E material but multiply the width by $n = \dfrac{E_2}{E_1}$.

Multiply stress by n for stiff material.



Stress Concentration (Bending)

$$\sigma_{\max} = K\left(\frac{Mc}{I}\right)_{\text{critical}}$$

Modulus of Rupture in Bending

$$R_B = \frac{M_U c}{I}$$

Plastic Bending

$$M_Y = \frac{I}{c}\sigma_Y$$

$0 \leq |y| \leq y_Y$: $\quad \sigma = -\dfrac{\sigma_Y}{y_Y}y$

$y_Y \leq |y| \leq c$: $\quad M = \dfrac{3}{2}M_Y\left(1 - \dfrac{y_Y^2}{3c^2}\right)$

$$M_P = \frac{3}{2}M_Y$$

$$M = \iint y\,\sigma\,dA = 2b\int_0^c \sigma\,y\,dy$$

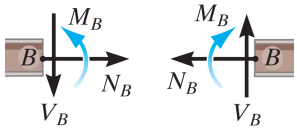
$M_e = \dfrac{\sigma_{\max} I}{c}$ (elastic)

$M_{e\text{-}p} = b\sigma_y\left(c^2 - \dfrac{1}{3}y_y^2\right)$ (elasto-plastic)

$M_p = b\sigma_y c^2$ (fully plastic)

$Z = \dfrac{M_p}{\sigma_Y}$ (plastic section modulus)

$k = \dfrac{M_p}{M_Y} = \dfrac{Z}{S}$ (shape factor)

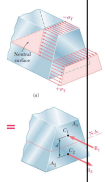


Unsymmetric Cross-Section (Plastic)

For non-symmetric sections, neutral axis $\neq$ centroidal axis.

Neutral axis divides cross-section into **equal areas**.

$$M_p = \left(\frac{1}{2}A\sigma_Y\right)d$$



d: distance between tension/compression centroids

Unsymmetric Bending

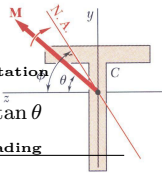
Resolve moment into principal axes:

$$M_z = M\cos\theta, M_y = M\sin\theta$$

Superpose stress distributions:

$$\sigma = -\frac{M_z y}{I_z} + \frac{M_y z}{I_y}$$

Neutral axis orientation



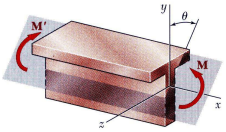
$$\tan\phi = \frac{y}{z} = \frac{I_z}{I_y}\tan\theta$$

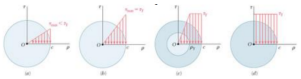
Eccentric Axial Loading

$$\sigma_x = \frac{P}{A} - \frac{M_z y}{I_z} + \frac{M_y z}{I_y}$$

Neutral axis:

$$\frac{M_z}{I_z}y - \frac{M_y}{I_y}z = \frac{P}{A}$$





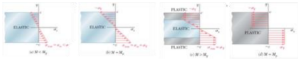
$\rho_{max} \leq \rho$: $\quad T = \dfrac{I\tau_Y}{c}$

$0 \leq \rho \leq \rho_Y$: $\quad \tau = \dfrac{\tau_Y}{\rho_Y}\rho$

$\rho_Y \leq \rho \leq c$: $\quad T = \dfrac{4}{3}T_Y\left(1 - \dfrac{1}{4}\dfrac{\rho_Y^3}{c^3}\right)$

$T = \dfrac{4}{3}T_Y\left(1 - \dfrac{1}{4}\dfrac{\phi_Y^3}{\phi^3}\right)$

$T_p = \dfrac{4}{3}T_Y$



$\sigma_{max} \leq \sigma_Y$: $\quad \sigma = \dfrac{Mc}{I}$

$0 \leq \sigma \leq \sigma_Y$: $\quad \sigma_x = -\dfrac{\sigma_Y}{y_Y}y$

$\sigma_Y \leq \sigma \leq c$: $\quad M = \dfrac{3}{2}M_Y\left(1 - \dfrac{1}{3}\dfrac{y_Y^2}{c^2}\right)$

$M = bc^2\sigma\left(1 - \dfrac{1}{3}\dfrac{y_Y^2}{c^2}\right)$

$M = \dfrac{3}{2}M_Y$

NEW – Residual Curvature

$$\frac{1}{\rho_{plastic}} = \frac{\varepsilon_{x,Yield}}{y_{Yield}}$$

$$\frac{1}{\rho_{unload}} = \frac{M}{EI}$$

$$\frac{1}{\rho_r} = \frac{1}{\rho_{plastic}} - \frac{1}{\rho_{unload}}$$

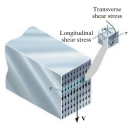
Elastic Curve

$$\frac{1}{\rho} = \frac{d^2y}{dx^2} = \frac{M}{EI}, \quad \frac{d^4y}{dx^4} = \frac{w}{EI}$$

$M > 0$: sagging
$M < 0$: hogging
Pin/Roller/Journal/Thrust support: $y = 0$, $y' \neq 0$
Fixed support: $y = 0$, $y' = 0$
Free End: $M = EIy'' = 0$, $V = EIy''' = 0$
Segments: $y_1(a) = y_2(a)$, $y_1'(a) = y_2'(a)$
Symmetry: $y(\text{center}) = y_{\max}$, $y'(\text{center}) = 0$
Complementary Coordinates:
$x_2 = L - x_1$
$y_1'(x_1 = a) = -y_2'(x_2 = L - a)$

Transverse and Longitudinal Shear

$$\tau = \frac{VQ}{It}$$



V: shear force, Q: first moment of area
I: second moment of inertia, t: width

Narrow Rectangular Beam

$$\tau_{xy} = \frac{3P}{2A}\left(1 - \frac{y^2}{c^2}\right)$$

Shear Flow

$$q = \frac{VQ}{I}$$

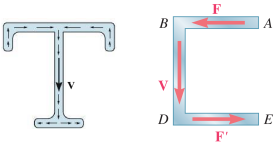
Force for One Nail

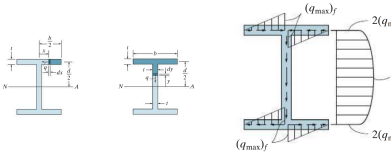
$$F_{\text{nail}} = \frac{q \cdot s}{n} \quad \left(\frac{\text{Force}}{\text{Length}} \cdot \frac{\text{Length}}{\text{Nail}} \div \text{rows} = \frac{\text{Force}}{\text{Nail}}\right)$$

s: spacing between nails
n: number of rows of nails

Unsymmetric Beam / Shear Center

$$F = \int_A^B q\,ds, \quad V = \int_B^D q\,ds, \quad e = \frac{Fh}{V}$$





Transformations of Plane Stress

Let $\sigma_{\text{avg}} = \dfrac{\sigma_x + \sigma_y}{2}$, $R = \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$$\sigma_{x'} = \sigma_{\text{avg}} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$
$$\sigma_{y'} = \sigma_{\text{avg}} - \frac{\sigma_x - \sigma_y}{2}\cos 2\theta - \tau_{xy}\sin 2\theta$$
$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}, \quad \sigma_{\max,\min} = \sigma_{\text{avg}} \pm R$$
$$\tan 2\theta_s = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}}, \quad \tau_{\max} = R, \quad \sigma' = \sigma_{\text{avg}}$$

θ_{p1} brings the x face to $\sigma_{\max}$

Transformations of Plane Strain

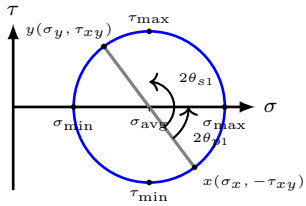
Let $\varepsilon_{\text{avg}} = \dfrac{\varepsilon_x + \varepsilon_y}{2}$, $R = \sqrt{\left(\dfrac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\dfrac{\gamma_{xy}}{2}\right)^2}$

$$\varepsilon_{x'} = \varepsilon_{\text{avg}} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$
$$\varepsilon_{y'} = \varepsilon_{\text{avg}} - \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta - \frac{\gamma_{xy}}{2}\sin 2\theta$$
$$\frac{\gamma_{x'y'}}{2} = -\frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta$$

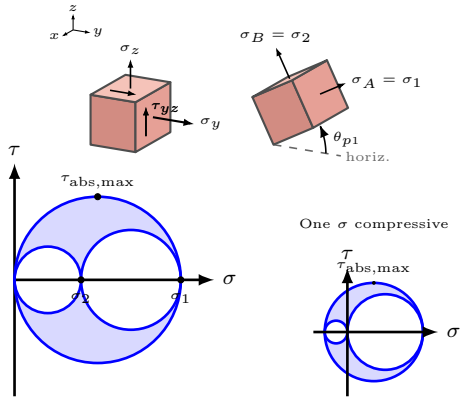
$$\tan 2\theta_p = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y}, \quad \varepsilon_{\max,\min} = \varepsilon_{\text{avg}} \pm R$$
$$\tan 2\theta_s = -\frac{\varepsilon_x - \varepsilon_y}{\gamma_{xy}}, \quad \frac{\gamma_{\max}}{2} = R, \quad \varepsilon' = \varepsilon_{\text{avg}}$$

θ_{p1} brings the x face to $\varepsilon_{\max}$

Mohr's Circle



Mohr's Circle — 3D



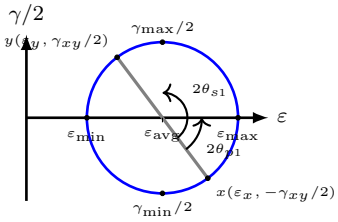
2D & 3D Shear Sign

(+): Face & arrow both positive or both negative direction
(-): Otherwise

Axis Substitution

Stress Plane	Points to Plot
(x, y)	$(\sigma_x, -\tau_{xy})$ (σ_y, τ_{xy})
(y, z)	$(\sigma_y, -\tau_{yz})$ (σ_z, τ_{yz})
(z, x)	$(\sigma_z, -\tau_{xz})$ (σ_x, τ_{xz})

Strain Transformations

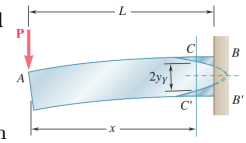


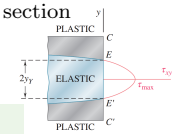
Plastic Deformation due to Shear

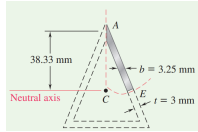
$Px = \frac{3}{2}M_Y\left(1 - \frac{y^2}{3c^2}\right)$
x: distance from free end

Shear Stress

$\tau_{xy} = \frac{3}{2}\frac{V}{A'}\left(1 - \frac{y^2}{y_Y^2}\right)$
A': area of elastic section
$A' = 2by_Y$ for rectangular cross section







Tilted cut
Shear at top is 0

Failure Theories

Ductile

Maximum Shearing Stress Criterion (Tresca)
$\tau_{\max} = \frac{1}{2}\sigma_Y$ (Tresca, uniaxial tension)
$\tau_{\max} = \frac{1}{2}|\sigma_1 - \sigma_2|$ (plane stress, general)
$\tau_{\max} = \frac{1}{2}\sigma_{\max}$ (plane stress, if $\sigma_{\min} = 0$)
If σ_a and σ_b have the same sign: $|\sigma_a| \leq \sigma_Y$ and $|\sigma_b| \leq \sigma_Y$ must hold.
If σ_a and σ_b have opposite signs: $|\sigma_a - \sigma_b| \leq \sigma_Y$ must hold.

Maximum-Distortion-Energy Criterion (von Mises)
$u_d = \frac{1}{6G}(\sigma_a^2 - \sigma_a\sigma_b + \sigma_b^2)$ (distortion energy per unit volume)
$\sigma_a^2 - \sigma_a\sigma_b + \sigma_b^2 \leq \sigma_Y^2$ (von Mises failure criterion)

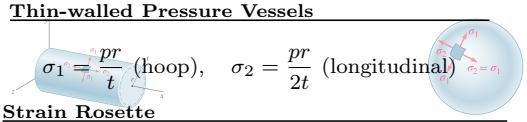
Brittle

Maximum Normal Stress Criterion
$|\sigma_a| \leq \sigma_U$ and $|\sigma_b| \leq \sigma_U$ (tension)

Maximum-Normal-Strain Criterion (Saint-Venant)
$|\epsilon_a| \leq \epsilon_U$ and $|\epsilon_b| \leq \epsilon_U$ (tension)

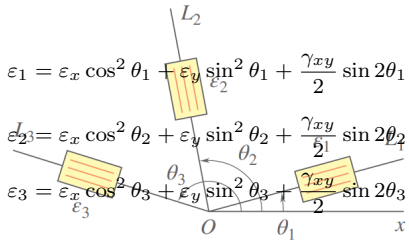
Mohr's Criterion
Draw Mohr's circles for UTS and UCS, state circle must fit inside circles.
Can add torsion test circle centered at origin with radius equal to τ_U.
Enveloppe or tangent line around circles.

Thin-walled Pressure Vessels



$\sigma_1 = \dfrac{pr}{t}$ (hoop), $\sigma_2 = \dfrac{pr}{2t}$ (longitudinal)

Strain Rosette



$$\varepsilon_1 = \varepsilon_x\cos^2\theta_1 + \varepsilon_y\sin^2\theta_1 + \frac{\gamma_{xy}}{2}\sin 2\theta_1$$
$$\varepsilon_2 = \varepsilon_x\cos^2\theta_2 + \varepsilon_y\sin^2\theta_2 + \frac{\gamma_{xy}}{2}\sin 2\theta_2$$
$$\varepsilon_3 = \varepsilon_x\cos^2\theta_3 + \varepsilon_y\sin^2\theta_3 + \frac{\gamma_{xy}}{2}\sin 2\theta_3$$

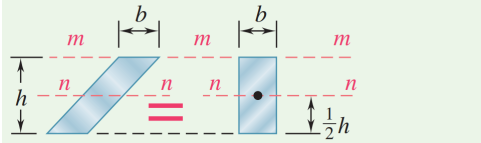
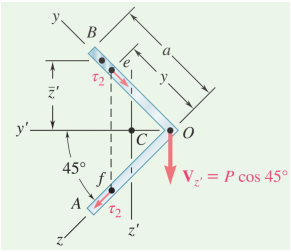


Fig. 2 Parallelogram and equivalent rectangle for determining moments of inertia.

same principle for centroid



Shearing Stresses Due to $V_{z'}$. Using Fig. 5, reconsider point e:

$$\bar{z}' = \tfrac{1}{2}(a + y)\cos 45°$$
$$Q = (a - y)t\bar{z}' = \tfrac{1}{2}(a^2 - y^2)t\cos 45°$$
$$\tau_2 = \frac{V_{z'}Q}{I_y t} = \frac{(P\cos 45°)[\frac{1}{2}(a^2 - y^2)t\cos 45°]}{(\frac{1}{3}ta^3)t} = \frac{3P(a^2 - y^2)}{4ta^3}$$